⌈Def. Define

$$f:[0,1]\to\mathbb{R}: x\mapsto \begin{cases} \frac{1}{p} & \text{if } x\in\mathbb{Q} \text{ with } x=\frac{q}{p}, \text{ irreducible form,} \\ 0 & \text{otherwise.} \end{cases}$$

Then,

1). f is continuous at every irrational numbers, and discontinuous at every rational numbers.

2) f is Riemann-integrable.

Proof. observe that for each $n\in\mathbb{N}$, the set

$$S_n=\{x\in[0,1] \mid f(x)>\tfrac{1}{n}\}=f^{-1}[(\tfrac{1}{n},\infty)]$$

is finite set. Let $x_0\in[0,1]$ be an irrational number.

Given $\varepsilon>0$, there is $N\in\mathbb{N}$ s.t $\frac{1}{N}<\varepsilon$. Since the S_N is finite, we can take $\delta>0$ s.t $\min\{|x_0-y| \mid y\in S_N\}$

$$x\in(x_0-\delta,x_0+\delta)\Rightarrow |f(x_0)-f(x)|=f(x)\le\frac{1}{N}<\varepsilon.$$

So, f is continuous at any irrational. And, for rational $\frac{q}{p}\in[0,1]$,

$$\text{for any } \delta>0, \text{ there is an irrational } x\in(\tfrac{q}{p}-\delta,\tfrac{q}{p}+\delta) \text{ s.t } |\tfrac{1}{p}-f(x)|=\tfrac{1}{p}>\tfrac{1}{2p}.$$

So it is not continuous at rational. ⌋ 1).

Given $\varepsilon>0$, there is $N\in\mathbb{N}$ s.t $\frac{1}{N}<\varepsilon$.

Denote $S_N=\{x_1,x_2,\cdots,x_k\}$. And, for $0<d<\min\{|x_i-x_j| \mid 1\le i<j\le k\}$.

Now, construct a partition as

$$P=\{x_1-\tfrac{d}{2},x_1+\tfrac{d}{2},x_2-\tfrac{d}{2},\cdots,x_k-\tfrac{d}{2},x_k+\tfrac{d}{2}\}\cap[0,1].$$

Then, for the upper sum $U(f,P)$ and $L(f,P)$,

$$U(f,P)-L(f,P)=U(f,P)<1\cdot\frac{1}{N}+dk<\varepsilon+dk$$

Since the k is fixed, letting $d\to 0$, we obtained the result.